

Secondary Examination, 2018

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 70

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) There are in all seven questions in this question paper.
- 2) All the questions are compulsory.
- 3) In the beginning of each, the number of parts to be attempted has been clearly mentioned.
- 4) Marks allotted to the questions are indicated against them.
- 5) Start from the first question and proceed to the last one.
- 6) Do not waste your time over a question you cannot solve

Q1. All questions are solute.

(a) Evaluate the HCF (Highest Common Factor) this expression $(x-2)(x+4)^2$ and $(x-2)^2$.

- (i) $(x-2)^4$
- (ii) $(x-2)^2$
- (iii) $(x-2)(x+4)^2$
- (iv) $(x-2)$

Sol.

The largest positive integer which divides two or more interger without any remainder is called Highest Common Factor (HCF) or Greatest Common Divisor or Greatest Common Factor.

Therefore,

This expression HCF is $(x-2)$.

Hence, the value of hcf $(x-2)$.

(b) Find the mode 12,20,21,29,12,29,24 and 29.

Sol.

The number which appears most often in a set of numbers.

We have the number are 12,20,21,29,12,29,24and 20.

Therefore, the value of mode given this expression is 29.

Hence, the value of mode is 29.

(c) Find the value of $\cos 15^\circ + \cos 105^\circ$.

(i) $\frac{1}{2}$

(ii) $\frac{1}{\sqrt{2}}$

(iii) $\frac{\sqrt{3}}{2}$

(iv) $\sqrt{\frac{3}{2}}$

Sol .

We have this expression $\cos 15^\circ + \cos 105^\circ$

$$\cos(45^\circ - 30^\circ) + \cos(60^\circ + 45^\circ)$$

We know that

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$= \cos(45^\circ - 30^\circ) + \cos(60^\circ + 45^\circ)$$

$$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ + \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} + 1 + 1 - \sqrt{3}}{2\sqrt{2}}$$

$$= \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}$$

Hence, the value is $\frac{1}{\sqrt{2}}$

- (d) Find the second radius. Whose two center are intersect one another, the distance between the center of the circle is 3cm. if first circle radius of 2cm.

Sol.

We know that,

$$D = r_2 - r_1 \text{ (D= Center of distance)}$$

$$3 = r_2 - 2$$

$$r_2 = 3 + 2$$

$$r_2 = 5$$

Hence, the value of $r_2 = 5$.

- (e) Passing through the line diagonal in point is (a, a^2) and (b, b^2) .

(i) 1

(ii) $\frac{b}{a}$

(iii) $(a + b)$

(iv) $-(a - b)$

sol.

We have the point (a, a^2) and (b, b^2) .

We know that,

$$M = \frac{Y_2 - Y_1}{X_2 - X_1}$$

$$M = \frac{b^2 - a^2}{b - a}$$

$$M = \frac{(b - a)(b + a)}{(b - a)} = (b + a)$$

$$M = (a + b)$$

Hence, the value of $(a + b)$.

- (f) In figure. O is the center of circle. So PA and PB are tangent of circle whose touching point are A and B. if $\angle APB = 50^\circ$ so find the value of $\angle AOB$

Sol.

Let consider a circle centered at point O.

Let P be an external point from which two tangent PA and PB are drawn to the circle which are touching the circle at point A and B respectively.

AB is the segment joint point of contact A and B together such that it subtends $\angle AOB$ At center O of the circle.

Therefore,

The $\angle AOB$ is reflex angle of $\angle APB$

We have $\angle APB = 50^\circ$ since, $\angle AOB = 100^\circ$

Hence, the angle of $\angle AOB = 100^\circ$.

2. All question solve.

(a). find the line of equation. a line slope of b/a . and b intersect is positive side y-axis.

Sol.

$$M = \frac{b}{a}$$

$$y = b$$

$$x = 0$$

$$y - y_1 = m(x - x_1)$$

$$y - b = \frac{b}{a}(x - 0)$$

$$ay - ab = b$$

$$(ay - bx = ab)$$

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(b) Find the value of $\tan \frac{13\pi}{6}$.

Sol.

We have the expression is $\tan \frac{13\pi}{6}$

Subtract the 2π given this expression.

$$\tan \frac{13\pi}{6} - 2\pi$$

$$\tan \frac{13\pi - 12\pi}{6} = \frac{\pi}{6}$$

$$\tan \frac{\pi}{6} = 30^\circ$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

Hence, the value of $\frac{1}{\sqrt{3}}$.

(c). find the value of x. if decending order the evaluate the value are 48, 44, 41,36,(2x-8),(2x-6),14,11,8, of secondary.

Sol.

$$\begin{aligned}x &= \frac{48 + 44 + 41 + 36 + (2x + 8) + (2x - 6) + 14 + 11 + 8 + 6}{10} = 25 \\&= 210 + 4x = 250 \\&= 4x = 40 \\x &= 10\end{aligned}$$

(d).if $f(x) = \frac{x^2 - 4x + 13}{x + 1}$ such that find the value is $f(8)$.

Sol.

Simplify the expression $f(8)$.

$$\begin{aligned}f(8) &= \frac{(8)^2 - 4(8) + 13}{8 + 1} \\f(8) &= \frac{64 - 32 + 13}{9} = \frac{45}{9} \\f(8) &= 5\end{aligned}$$

Hence, the value of $f(8) = 5$.

3 All question are solute.

(a) Prove that

$$\frac{\cos(60^\circ - A) + \sin(30^\circ - A)}{\cos(60^\circ - A) - \sin(30^\circ - A)} = \cot A$$

Sol.

$$\text{LHS } \frac{\cos(60^\circ - A) + \sin(30^\circ - A)}{\cos(60^\circ - A) - \sin(30^\circ - A)}$$

$$= \frac{\cos(90^\circ - (60^\circ + A)) + \sin(30^\circ - A)}{\cos(90^\circ - (60^\circ - A)) - \sin(30^\circ - A)}$$

$$= \frac{\cos(30^\circ + A) + \sin(30^\circ - A)}{\sin(30^\circ + A) - \sin(30^\circ - A)}$$

We know that,

$$\sin C + \sin D = 2 \sin \frac{(C + D)}{2} \cos \frac{C - D}{2}$$

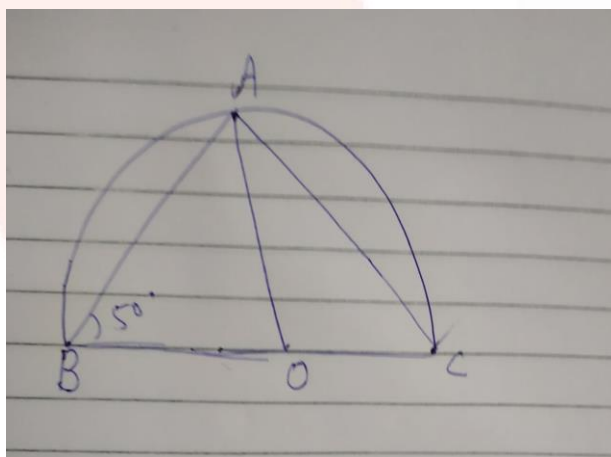
$$\sin C - \sin D = 2 \sin \frac{(C + D)}{2} \cos \frac{C - D}{2}$$

$$= \frac{2 \sin \frac{(30^\circ + A + 30^\circ - A)}{2} \cos \frac{(30^\circ + A - 30^\circ + A)}{2}}{2 \cos \frac{(30^\circ + A + 30^\circ - A)}{2} \sin \frac{(30^\circ + A - 30^\circ + A)}{2}}$$

$$= \frac{2 \sin 30^\circ \cos A}{2 \sin 30^\circ \sin A} = \cot A$$

(b). figure in, find the angle $\angle OAC$.whose hemisphere of center is O and diameter is BC. If the angle $\angle ABO = 50^\circ$.

Sol.



We know that hemisphere corner are right angle.

Then,

$$OA = OB \text{ (radius)}$$

Same angle of side are equal angle .

$$\angle OBA = \angle OAB = 50^\circ$$

$$\begin{aligned}\text{Since, } \angle OAC &= \angle BAC - \angle OAB \\ &= 90^\circ - 50^\circ = 40^\circ\end{aligned}$$

Hence, the angle of $\angle OAC = 40^\circ$.

(c). find the intersection by the y axis.. Therefore have a line of $x + y = 9$ a tilt positive x axis.

Sol.

$$x + y = 9$$

$$M = \frac{-b}{a} = -\frac{1}{1}$$

$$M = -1$$

$$\tan \theta = \tan 135^\circ$$

$$\theta = 135^\circ$$

Simplify further,

$$x + y = 9$$

$$\frac{x}{9} + \frac{y}{9} = \frac{9}{9}$$

$$\frac{x}{9} + \frac{y}{9} = 1$$

Then, $y = 9$

Hence, the value of $y = 9$.

(d) Find the height of a cylinder. Whose curved surface area is 900π and its radius 10cm.

Sol.

Simplify the height of cylinder.

We know that

The curved surface area of cylinder is $2\pi rh$.

$$2\pi rh = 900\pi$$

$$2\pi \cdot 10 \cdot h = 900\pi$$

$$2 \cdot 10 \cdot h = 900$$

$$h = 45\text{cm}$$

Hence, the value of $h = 45\text{cm}$.

4 Q. All the question are solute.

(a) Find the selling rate. Sanjay buy a cycle of 3,528 . Whose including a selling price.
If the actual price of cycle is 3,360 .

Sol.

Simplify the selling rate of cycle.

We have buy price of cycle is 3,528 and actual price is 3,360 .

We simplify the extra charge of his actual cycle rate is .

$$3528 - 3360 = 168$$

We know that general forumula of profit rate.

$$\text{sellingrate} = \frac{\text{difference of buying price - difference of actual price}}{\text{actual price}} \cdot 100$$

$$= \frac{168}{3360} \times 100$$

$$= 5\%$$

Hence, the selling rate of cycle is 5%.

(b). prove that

$$\cot A - \tan A = 2 \cot 2A.$$

Sol.

Simplify, we have the expression is $\cot A - \tan A = 2 \cot 2A$.

$$\cot A - \tan A = 2 \cot 2A$$

We take LHS

$$\cot A - \tan A = \frac{\cos A}{\sin A} - \frac{\sin A}{\cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{\sin A \cdot \cos A}$$

We know that,

$$\cos^2 A - \sin^2 A = \cos 2A$$

Simplify further,

$$= \frac{\cos 2A}{\sin A \cos A}$$

Multiply of 2 numerator and denominator.

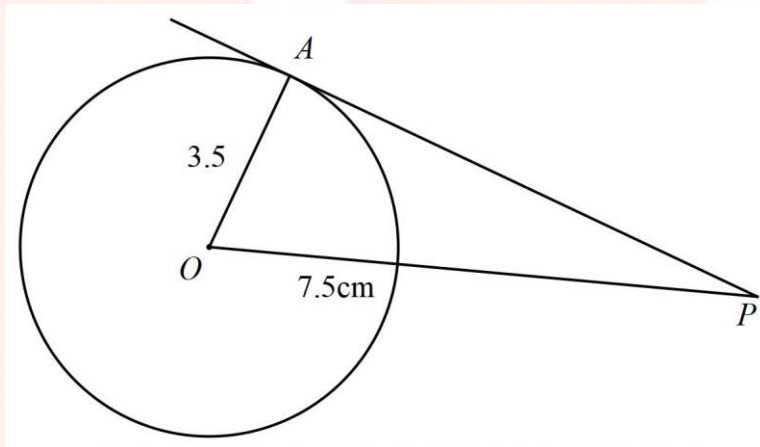
$$\begin{aligned}
 &= \frac{\cos 2A}{\sin A \cos A} \\
 &= \frac{2 \cos 2A}{2 \sin A \cos A} \quad (2 \sin a \cos a = \sin 2a) \\
 &= \frac{2 \cos 2A}{\sin 2A} = 2 \cot 2A \\
 &= 2 \cot 2A \text{ RHS} \\
 &\text{Hence, proved LHS=RHS.}
 \end{aligned}$$

(c). Draw a circle having radius 3.5cm. also draw a tangent from the point 7.5cm To center.

Sol.

Draw in figure,

We have a radius is OA is 3.5 and OP is 7.5 .



Hence, draw above in figure .

(d). find curved surface area of cone. Whose given the cone of height and radius are 15cm and 8cm.

Sol.

Simplify the curved surface area of cone.

We know that the formula is curved surface area of cone is πrl .

$$\pi rl$$

$$l = \sqrt{15^2 + 8^2}$$

$$l = \sqrt{225 + 64} = \sqrt{289}$$

$$l = 17$$

We using the formula is πrl .

$$\pi rl$$

$$= 3.14 \times 8 \times 17 = 427.04 \text{ cm}$$

Hence, the value of curved surface area of cone is 427.04cm .

5. All question are solute.

(a). find the LCM(Lowest Common Multiple) $x^2 + x - 12$, $x^2 - 9$, and $x^2 + 8x + 16$.

Sol.

The smallest positive number that is a multiply of two or more numbers.

$$x^2 + x - 12 = x^2 + 4x - 3x - 12$$

$$= x(x + 4) - 3(x + 4)$$

$$= (x + 4)(x - 3)$$

$$x^2 - 9 = (x + 3)(x - 3)$$

$$x^2 + 8x + 16 = x^2 + 4x + 4x + 16$$

$$= x(x + 4) + 4(x + 4)$$

$$= (x + 4)(x + 4)$$

We get the value LCM is $(x + 4)^2 (x - 3)(x + 3)$.

Hence, the value of $(x - 3)(x + 4)$.

(b). Prove that.

$$\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = 1 + \sec A \operatorname{cosec} A.$$

Sol.

We take the RHS

$$= \frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$$

$$\begin{aligned}
 &= \frac{\tan A}{1 - \frac{1}{\tan A}} + \frac{\cot A}{1 - \tan A} \\
 &= \frac{\tan^2 A}{1 - \tan A} - \frac{\cot A}{\tan A - 1} \\
 &= \frac{\tan^2 A - \cot A}{\tan A - 1} \\
 &= \frac{\frac{\sin^2 A}{\cos^2 A} - \frac{\cos A}{\sin A}}{\frac{\sin A}{\cos A} - 1} \\
 &= \frac{\frac{\sin^3 A - \cos^3 A}{\sin A \cos A (\sin A - \cos A)}}{\frac{(\sin^2 + \cos^2)(\sin A - \cos A) + \cos A \sin^2 A - \sin A \cos A}{\sin A \cos A (\sin A - \cos A)}} \\
 &= \frac{(\sin^2 + \cos^2)(\sin A - \cos A) + \sin A \cos A (\sin A - \cos A)}{\sin A \cos A (\sin A - \cos A)} \\
 &= \frac{(\sin^2 + \cos^2) + \sin A \cos A}{\sin A \cos A} \\
 &= \frac{(\sin^2 + \cos^2)}{\sin A \cos A} + 1 \\
 &= 1 + \tan A + \cot A \\
 &= 1 + \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} \\
 &= 1 + \frac{1}{\sin A \cos A} \\
 &= 1 + \operatorname{cosec} A \cdot \sec A \quad \text{RHS}
 \end{aligned}$$

Hence, proved.

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(c). Determine the ratio in which the line $3x - 5y = 4$ divides the line segment joining the points (6,-4) and (4, 4).

Sol.

Suppose the line $3x + y - 9 = 0$ divide the join of A(6, -4) and B(4, 4) in the ratio k:1 at the point P.

Now, the coordinate of P are

$$p \left[\frac{4k+6}{k+6} \right], \left[\frac{4k-4}{k+6} \right]$$

Putting the value of equation $3x - 5y = 4$

$$3 \left[\frac{4k+6}{k+6} \right] - 5 \left[\frac{4k-4}{k+6} \right] - 4$$

$$12k + 18 - 20k + 20 - 4k - 24 = 0$$

$$-12k + 14 = 0$$

$$12k = 14$$

$$k = \frac{14}{12} = \frac{7}{6}$$

$$= \frac{7}{6} : 1$$

$$= 7 : 6$$

Hence the value of 7 : 6

d). A man of yearly income of 2016-2017. (excluding house rent). He deposits its 8000 per month in usual future fund account. He deposits 80,000 in his PDF account. He deposits 80,000 in his PDF account. Calculate the income tax payable by x while the maximum limit of saving is one and half lakh. Income tax rates are as follow:

Income	Income tax
Up to Rs. 2,50	zero
Rs 2,50,001 to 5,00,000	Rs 2,50,000 increase income of 10%

Education extra income tax 3%.

Sol.

Total income 6,20,000 .

Saving income $8000 \times 12 = \text{Rs } 96,000$.

P.F = Rs 80,000 .

Discount = $96,000 + 80,000 + 150,000$

= Rs 3,26,000

$620,000 - 326,000 = 294,000$.

Rate of income tax.

$$= 294,000 - 250,000 = 44,000$$

$$= 44,000 \times 10\%$$

$$= 44,000 \times \frac{10}{100} = 4400$$

$$= 4400 \times 3\%$$

$$= 4400 \times \frac{3}{100} = 132$$

Pay education of income tax = $4400 + 132 = \text{Rs } 4532$

Hence the value of income tax of education $\text{Rs } 4532$.

6.(a). find the median from the following frequency distribution.

Class interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency s	12	18	24	15	11

Sol.

We know that $x_1 = \frac{0+12}{2} = 6$ similarly $x_2, x_3, \dots, x_5$

And, $\sum f_1 x_1 = f_1 \times x_2$

C.I	f_1	x_1	$f_1 x_1$
0-10	12	5	60
10-20	18	15	270
20-30	24	25	600
30-40	15	35	575
40-50	11	45	495
	$\sum f_1 = 50$		$\sum f_1 x_1 = 1950$

$$\text{mean} = \frac{1950}{50} = 39$$

Hence, the value is mean 39 .

(b). A toy is the form of a cone mounted on a hemisphere of radius 3.5cm. The total height of the toy is 15.5cm. find the total surface area .

Sol

We have radius of hemisphere is 3.5cm.

Height of cone is $15.5 - 3.5 = 12\text{cm}$.

we get the slant height of cone,

$$l = \sqrt{r^2 + h^2}$$

$$= \sqrt{\left(\frac{7}{2}\right)^2 + 12^2} \text{ cm} = \frac{25}{2} \text{ cm}.$$

The total surface area of the toy = (curved surface of the hemisphere + curved surface of the cone).

$$(2\pi r^2 + \pi r l) \text{ sq units}$$

$$= (\pi \times r \times (2r + l)) \text{ cm}^2$$

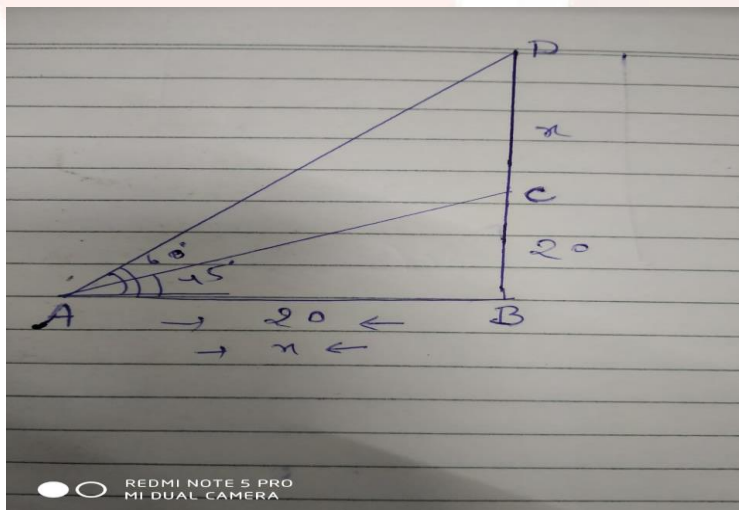
$$= \left\{ \frac{22}{7} \times \frac{7}{2} \times \left(7 + \frac{25}{2} \right) \right\} \text{ cm}^2$$

$$= \frac{429}{2} \text{ cm}^2 = 214.5 \text{ cm}^2$$

Hence, the total surface area of the toy is 214.5 cm^2 .

(c). The angle of elevation from the point to the top and end of tower which is on the building having height 20 m is 45° and 60° .

Sol.



we have the angle,

$\triangle BAC$

$$\tan 45^\circ = \frac{BC}{AB} = \frac{20}{x}$$

$$1 = \frac{20}{x}$$

$$x = 20$$

And, take angle

$\triangle BAD$

$$\tan 60^\circ = \frac{BD}{AB}$$

$$\tan 60^\circ = \frac{20 + x}{20}$$

$$\sqrt{3} = \frac{20 + x}{20}$$

$$20 + x = 20\sqrt{3}$$

$$x = 20\sqrt{3} - 20$$

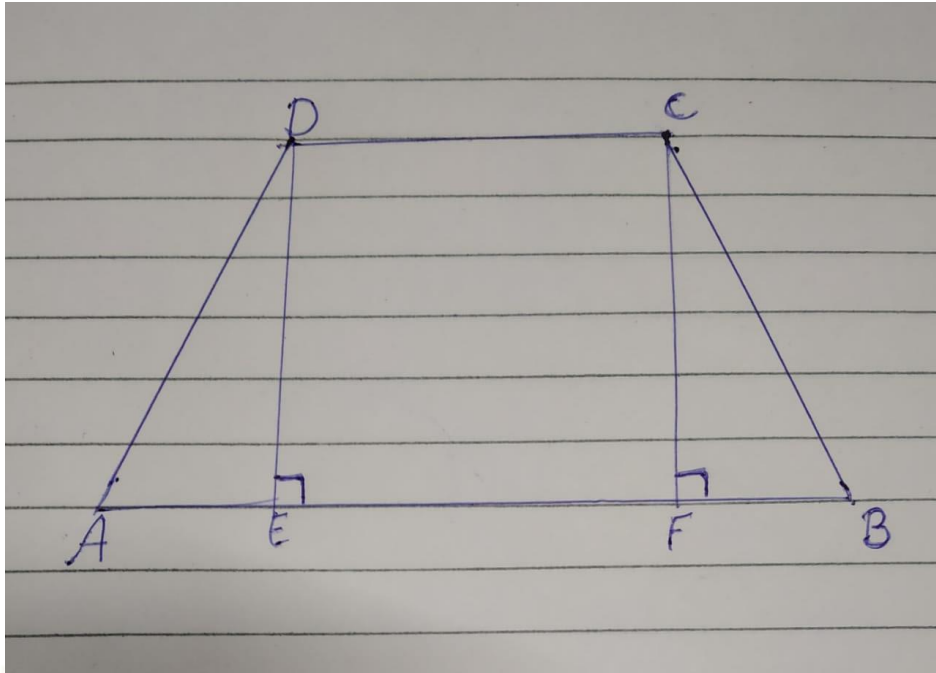
$$x = 20(\sqrt{3} - 1)$$

Hence, the value is $20(\sqrt{3} - 1)$

7. All question are solute.

(a) prove that if non parallel sides of trapezium are equal than prove that it is cyclic.

Sol.



Prove the expression.

In $\triangle AED$ and $\triangle BFC$, $AD=BC$
(Given) $DE=CF$ (distance between parallel sides is same)

$$\angle AED = \angle BFC = 90^\circ \quad \triangle AED \cong \triangle BFC \quad (\text{RHS Congruence criterion})$$

Since $\angle DAE = \angle CBF$ (CPCT).....(1) Since $AB \parallel CD$, AD is transversal

$\angle DAE + \angle ADC = 180^\circ$ (Sum of adjacent interior angles is supplementary).

$$\angle CBF + \angle ADC = 180^\circ \quad [\text{from (1)}]$$

Since sum of opposite angles is supplementary in trapezium ABCD is a cyclic trapezium.

Hence, proved .

Or

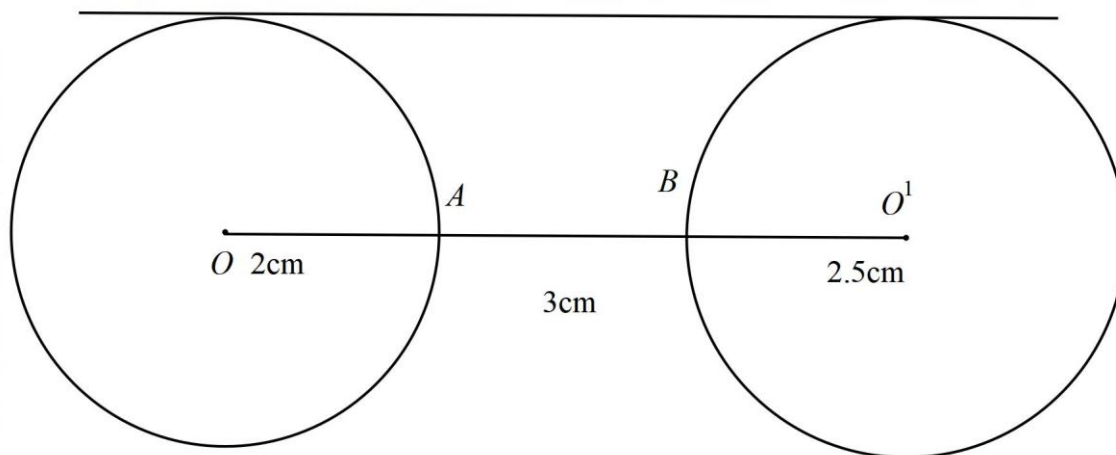
Q). Draw in two circle their radius are 2 cm and 2.5 cm with distance between of center whose center 7.5cm centimeters. Compose the common diagonal tangent of these verses and measure of length. Confirm calculation.

Sol.

We have two circle whose radius is 2cm and 2.5cm . their distance are each other 7.5cm.

We draw the figure below in.

We get a $OA=2\text{cm}$ and $O'B=2.5\text{cm}$ and $OO'=7.5\text{cm}$.



- (i) Find the quadratic equation $3x^2 - 6x + 2 = 0$.

Sol.

$$3x^2 - 6x + 2 = 0$$

We know that quadratic equation is,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \cdot 3 \cdot 2}}{2 \cdot 3}$$

$$x = \frac{6 \pm \sqrt{36 - 24}}{6}$$

$$x = \frac{6 \pm \sqrt{12}}{6} = \frac{6 \pm 2\sqrt{3}}{6}$$

$$x = \frac{6 \pm 2\sqrt{3}}{6}$$

Hence, the value of quadratic equation $x = \frac{6 \pm 2\sqrt{3}}{6}$.

- (b) The diagonal of the rectangular field is 30 m more than the shortest side of the field. If the larger side is 15m more than the shortest side. Then find the sides of the rectangular field.

Sol.

Let the small side of x.

we get larger side is $x+15$.

And diagonal is $x+30$

Now, simplify the expression.

We know that the rectangular diagonal is.

$$D = \sqrt{l^2 + b^2}$$

$$30 + x = \sqrt{(15 + x)^2 + x^2}$$

$$(30 + x)^2 = (15 + x)^2 + x^2$$

$$900 + 60x + x^2 = 225 + x^2 + x^2$$

$$x^2 + 30 - 60x + 225 - 900$$

$$x^2 - 30x - 625 = 0$$

$$x^2 + 15x - 45x - 675 = 0$$

$$x(x - 45) - 45(x + 15)$$

$$= (x - 45)(x + 15)$$

$$x = 45$$

$$x = -15$$

We get the value of x is $x = 45$.

Since, the small side is $x = 45$ and largest side is $x = 60$.

- (ii) Find the value of α & β $\alpha^3 + \beta^3$. If quadratic equation $ax^2 + bx + c = 0$ root point of α & β .

Sol.

Quadratic equation $ax^2 + bx + c = 0$. So, the original point α & β .

We know that,

$$\text{The sum of root } \alpha + \beta = -\frac{b}{a}.$$

$$\text{The product of root } \alpha \cdot \beta = \frac{c}{a}.$$

We given the function is $\alpha^3 + \beta^3$.

We know that the formula is $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta)$.

$$= (\alpha + \beta) \left[(\alpha + \beta^2) - \alpha\beta - 2\alpha\beta \right]$$

$$= \left(\frac{-b}{a} \right) \left[\left(\frac{-b}{b} \right)^2 - 3\alpha\beta \right]$$

$$= \left(\frac{-b}{a} \right) \left[\left(\frac{-b}{b} \right)^2 - 3 \left(\frac{c}{a} \right) \right]$$

$$= \left(\frac{-b}{a} \right) \left[\frac{b^2}{a^2} - \frac{3c}{a} \right]$$

$$= \left(\frac{-b}{a} \right) \left[\frac{b^2 - 3ca}{a^2} \right]$$

$$= \frac{-b^2 + 3abc}{a^3}$$

Hence, the value of $\frac{-b^2 + 3abc}{a^3}$.